

Data Analysis

FACTORS INFLUENCING UBER RIDE PRICES

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OVERVIEW

What Factors impacts?

Our analysis shows that Uber fares are positively correlated with both trip **distance** and **duration**, order located distribution and the peak demand occurs during typical rush **hours**.

Measures

Data Clean

Bar Chart

Order per Hour

Heat Map

Order Distribution

Scatter Plot

Duration vs Fare

Scatter Plot

Distance vs Fare

Data Visualization

Tools



pandas

matplotlib

seaborn

Folium

NumPy

scikit learn

kaggle

Data Collection

- **30195 rows * 19 columns**
- **Year 2021**
- **Sydney**



THE WAY WE SOLVE

Motivation

Uber has become a popular choice for urban travel in Sydney, understanding its **pricing mechanism** is valuable for passengers. Passengers are often not sure about what **influences the final fare**.

Problems

- **Peak demand periods?**
- **Concentrated locations?**
- **Fare prices? Distance? Duration?**

Metrics

- Order count per 2 hours
- Pickup suburb frequency
- Test pricing factors

Assumptions

- Only consider **valid distance, positive fare and duration (≥ 5 minutes and ≤ 60 minutes)**.
- **Uber vehicle types** are not considered. (UberX, UberXL, Uber Travel, etc).
- Our Data from Kaggle can show the **real Uber activity in Sydney**.



DATA CLEAN AND SELECTION

1 Data Selection

11 out of **19** columns: remove unnecessary data (Trip id, Pickup State, Pickup Country, etc.)

2 Missing value

368 missing value in **pickup/drop-off suburb - dropna**

3 Data range

Positive distance and fare, duration (≥ 5 minutes and ≤ 60 minutes).

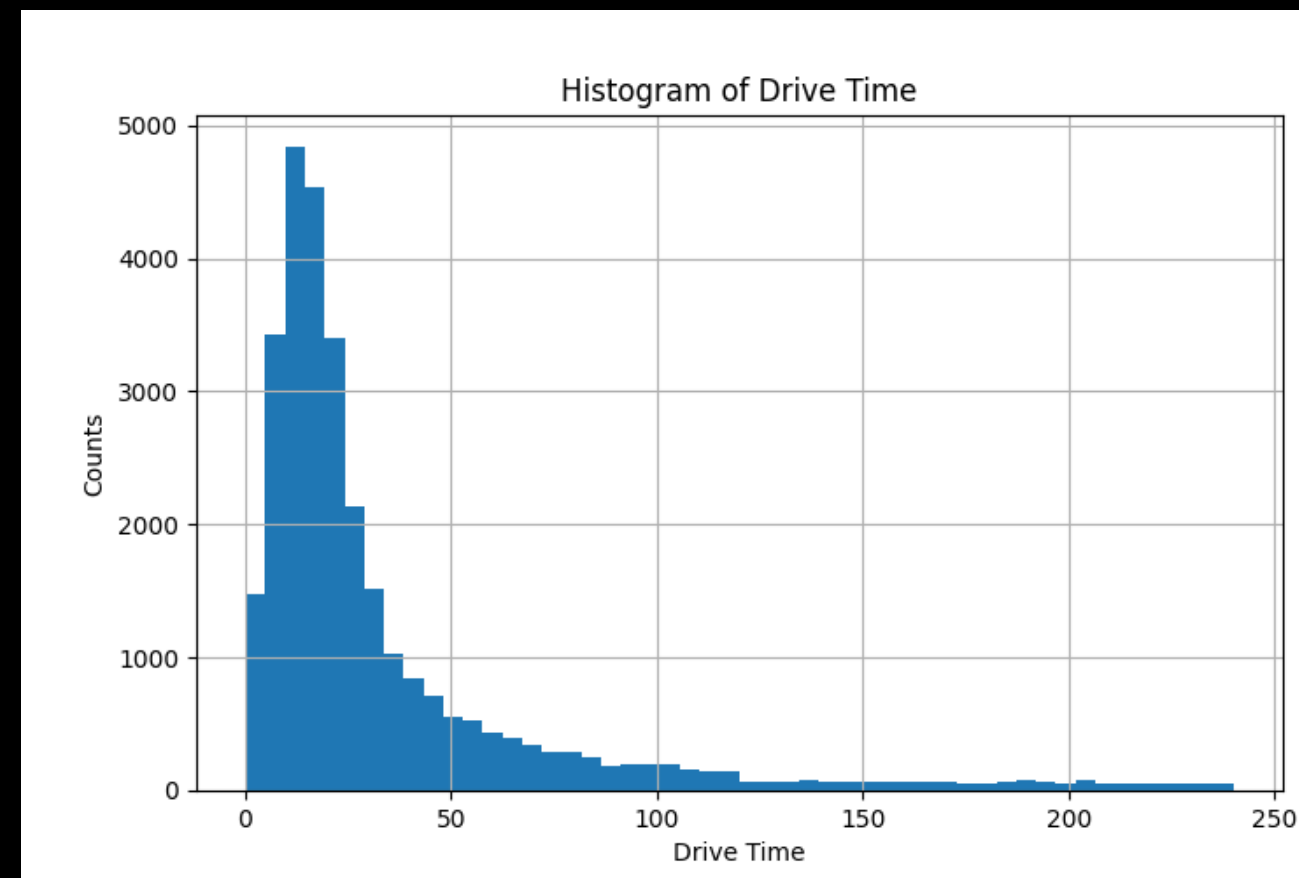
	Drive Time	Drive Distance Km	Uber Fare
count	29799.000000	29799.000000	29799.000000
mean	35.087957	11.406129	51.855672
std	41.204002	11.299843	48.928728
min	0.160000	0.016093	9.000000
25%	12.320000	5.085527	21.570000
50%	20.100000	8.931859	35.360000
75%	38.405000	13.727704	60.165000
max	239.970000	108.196197	272.340000



	Drive Time	Drive Distance Km	Uber Fare
count	23717.000000	23717.000000	23717.000000
mean	21.585866	10.217706	36.423282
std	12.521835	7.059761	20.304915
min	5.000000	0.128748	9.000000
25%	12.300000	5.278648	21.560000
50%	18.260000	8.851392	32.330000
75%	27.500000	13.293181	46.670000
max	60.000000	63.327686	149.890000

Limitations

- Data not included **Vehicle Type**, limiting analysis on price influencing factor.
- **Remove missing and extreme values** may slightly effect real situation



HOW DIFFERENT TIMES OF DAY AFFECT UBER FARE

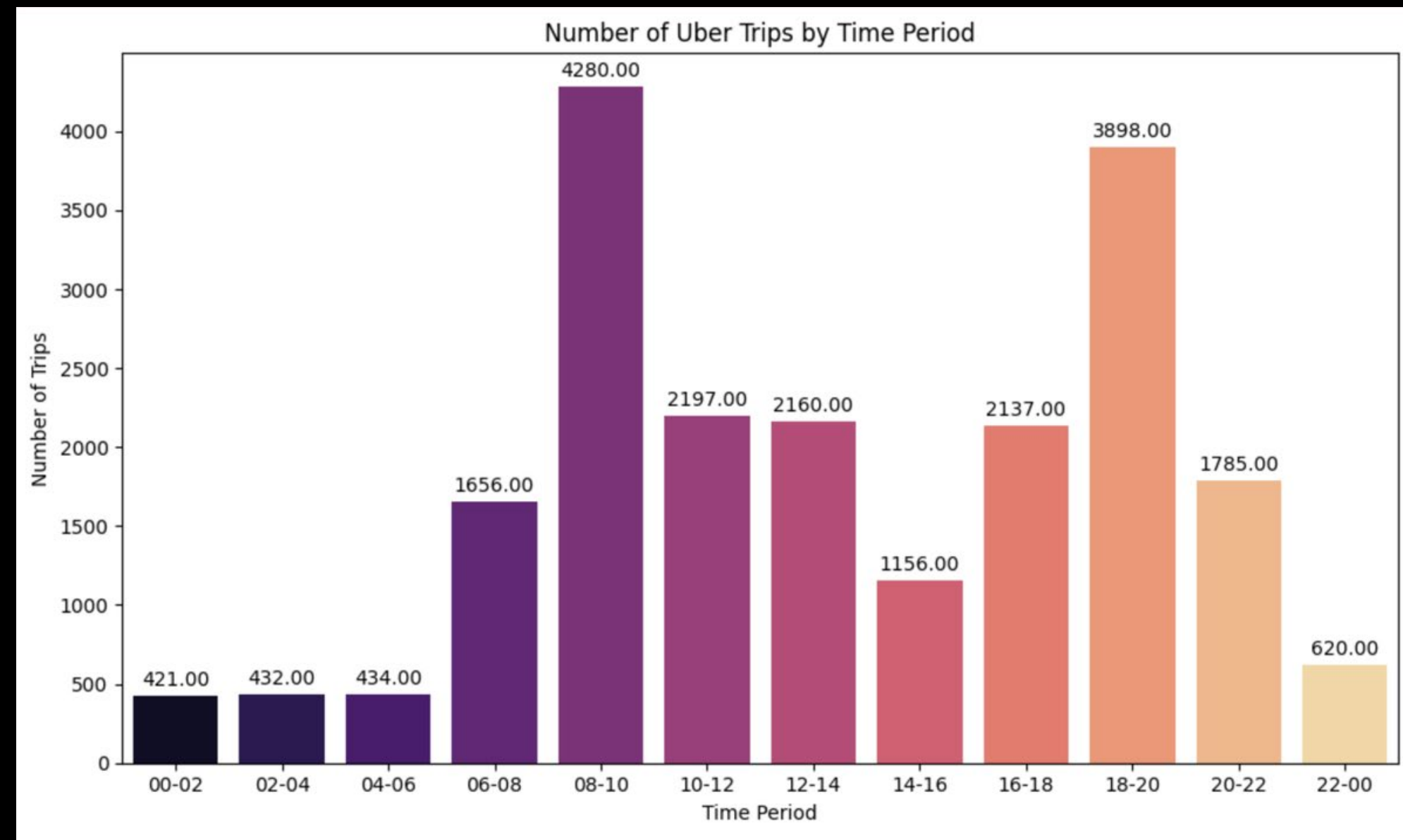
● with bar chart

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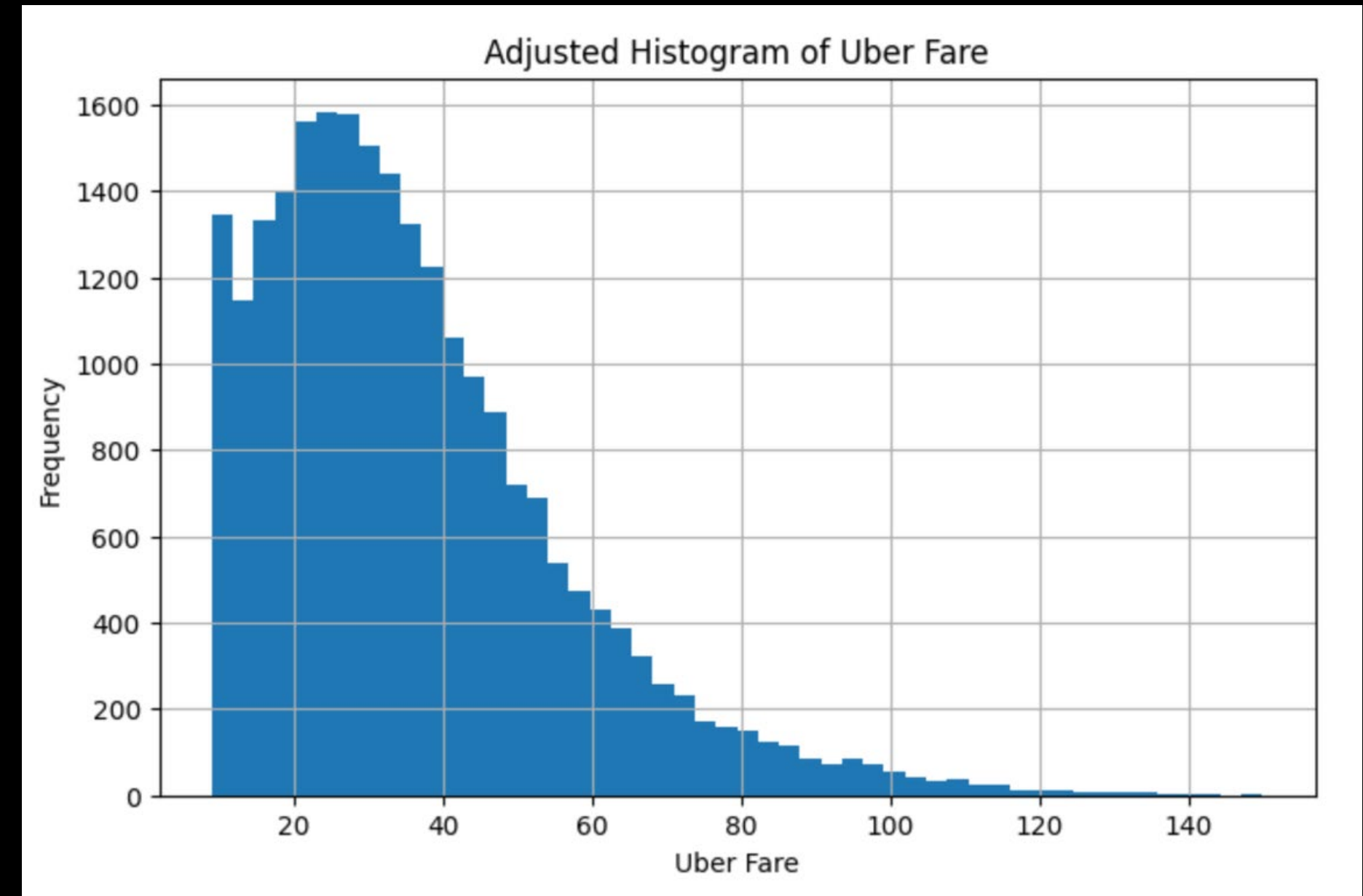
TRIP VOLUME BY TIME PERIOD

- All orders were grouped into 2 hour periods
- Observation: Most trips occur at:
8:00–10:00 AM
6:00–8:00 PM
- Fewer trips between 10:00 PM and early morning.



WHY USE MEDIAN INSTEAD OF MEAN

- Fare distribution is right -skewed:
most rides cluster at lower prices,
while a few rides are much more
expensive.
- Mean is easily affected by extreme
values (e.g., long or expensive
rides).
- The median represents the 50th
percentile and better reflects what
most users pay.

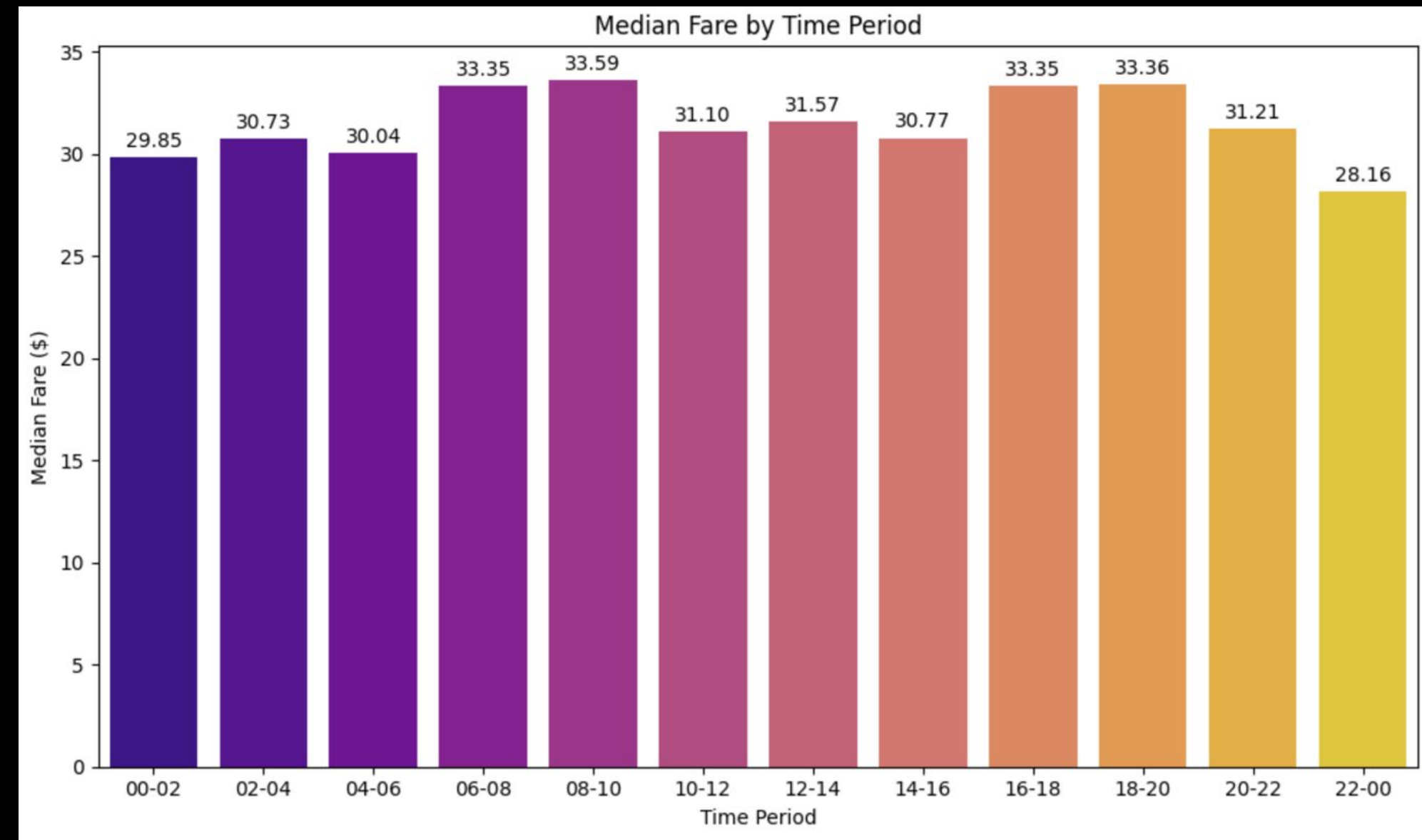
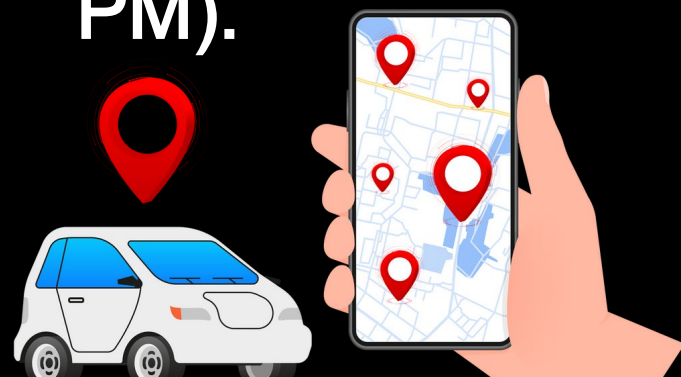


MEDIAN FARE BY TIME PERIOD

- Median fare calculated for each time slot and labels displayed above each bar for clarity.

- School Zone and Peak Hour Effect:

- 6–10 AM and 4 –8 PM have higher fares.
- Aligns with school zone (7:00 9:30 AM) and commute hours (4 PM).



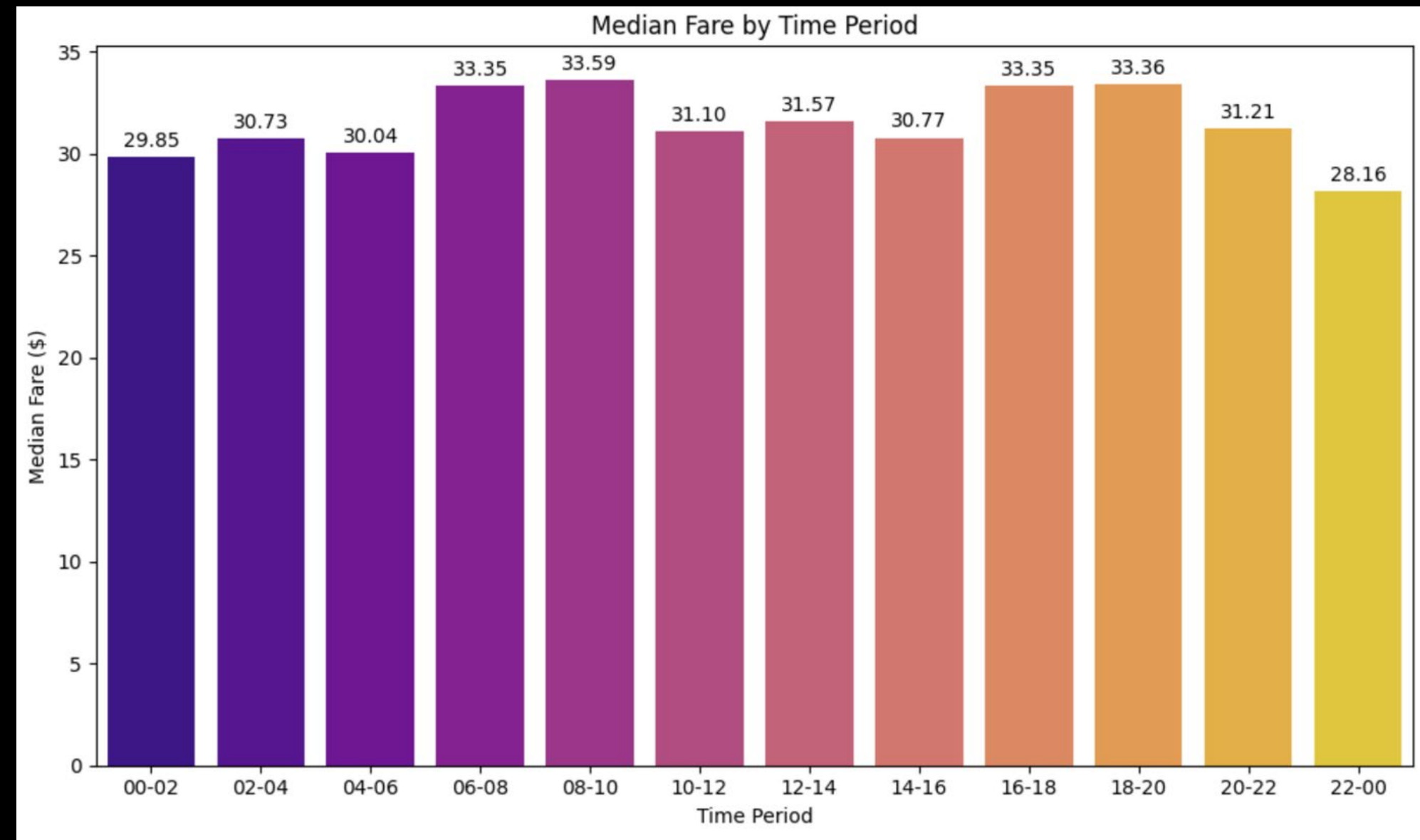
MEDIAN FARE BY TIME PERIOD

Low-Fare Periods

- 10:00 PM – 2:00 AM shows lower and more stable fares.

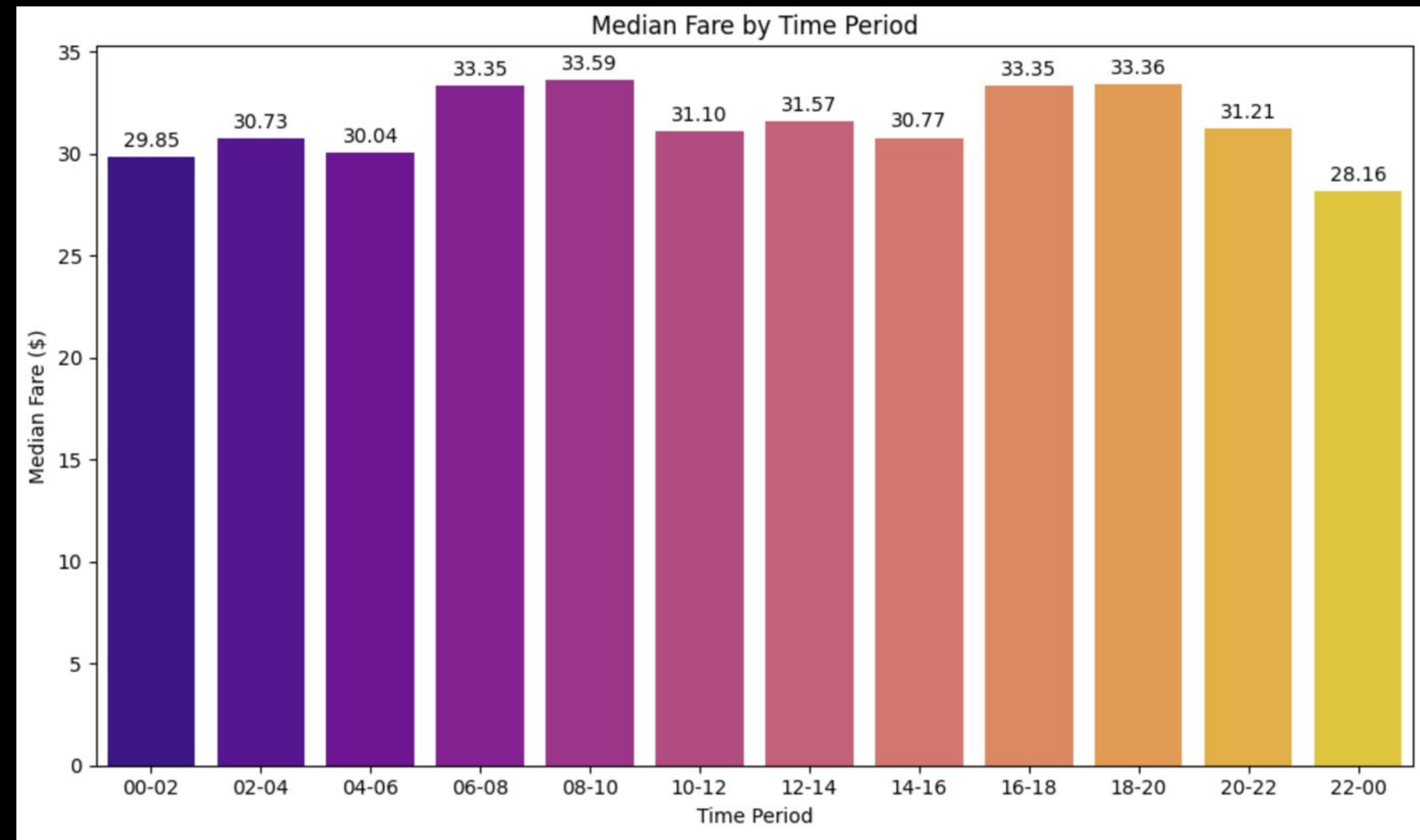
Possible reasons:

- Balanced or low supply-demand.
- Reduced traffic congestion.



CONCLUSION

- Uber prices in Sydney vary significantly with time of day.
- Higher fares during high demand hours: school zone & peak commuting times.
- Lower fares late at night when demand is low.
- Advice: Riders can save money by avoiding peak times.



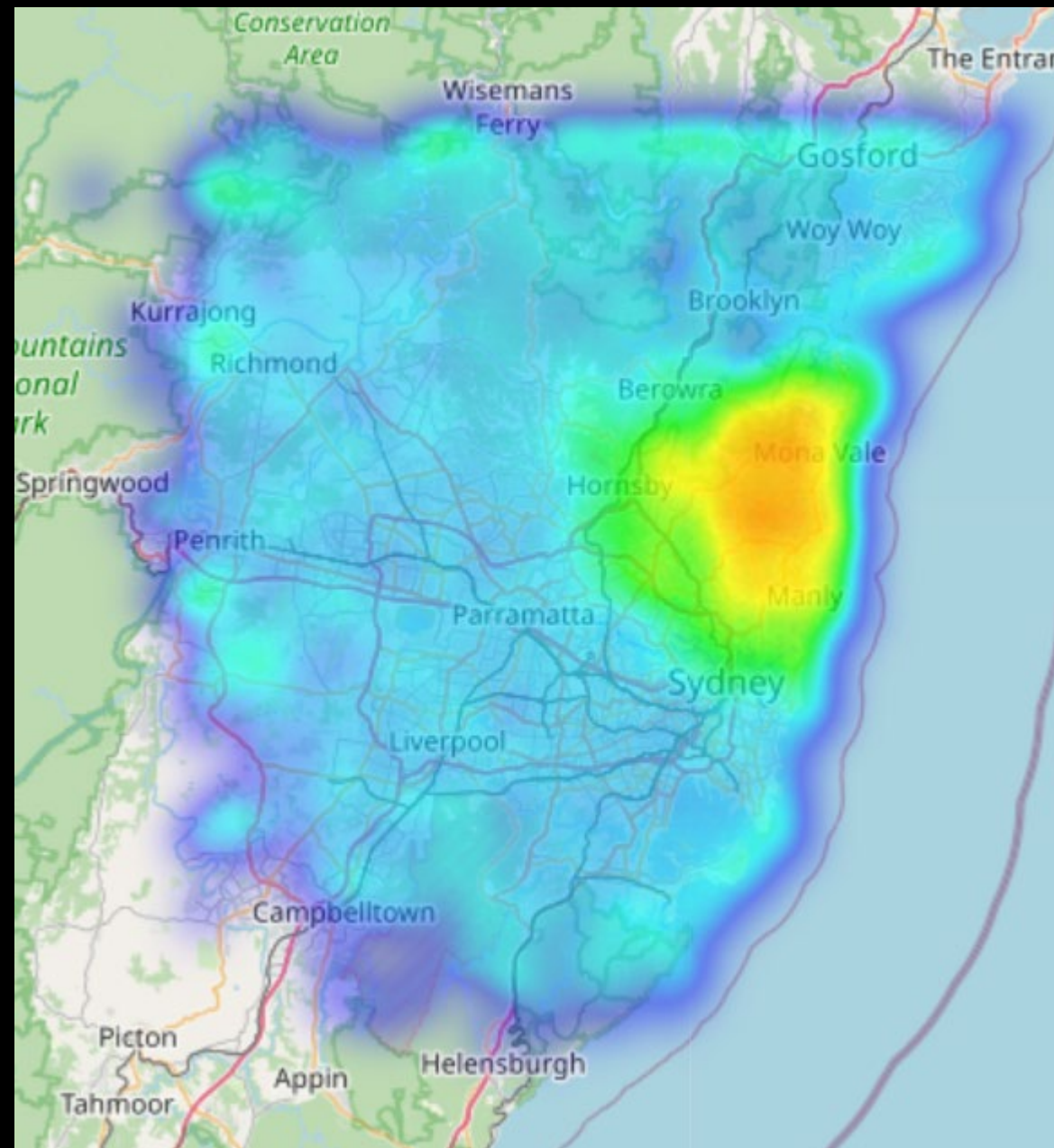
HOW GEOGRAPHIC LOCATION AFFECT UBER FARE

● with heat map and bar chart

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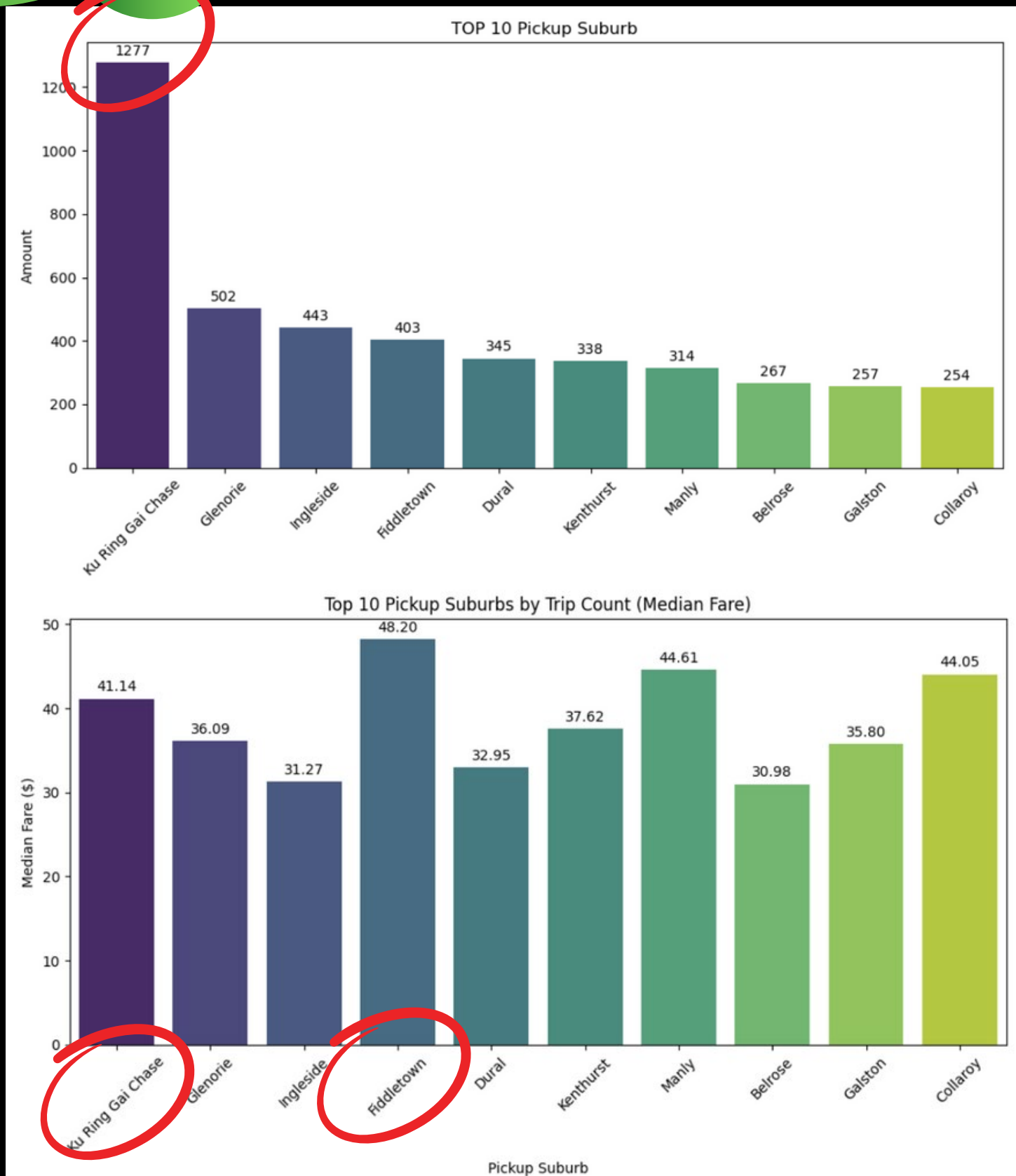


PICK-UP HEAT MAP



The highest concentration of Uber rides is in Sydney's city center and nearby areas, while remote regions have significantly fewer trips.

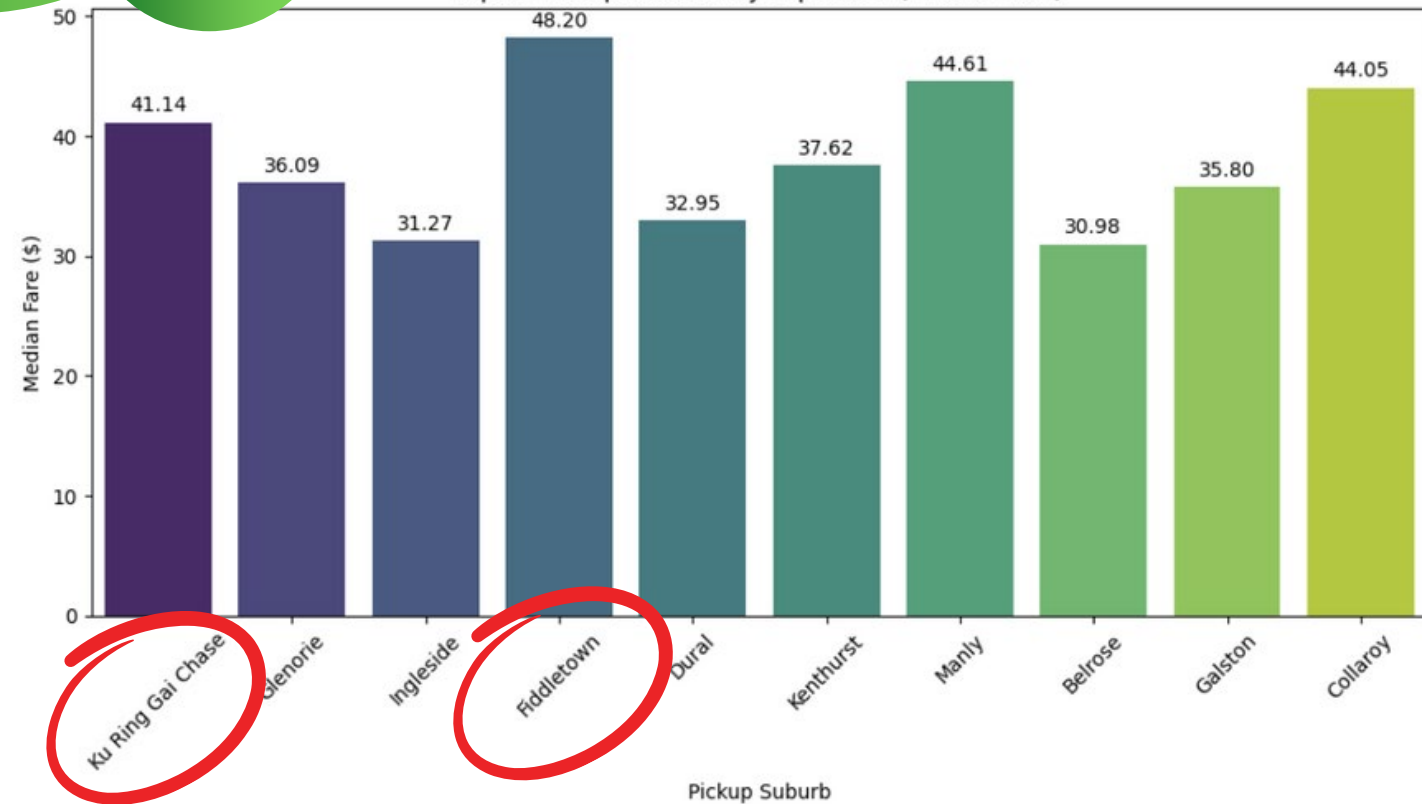
BAR CHART



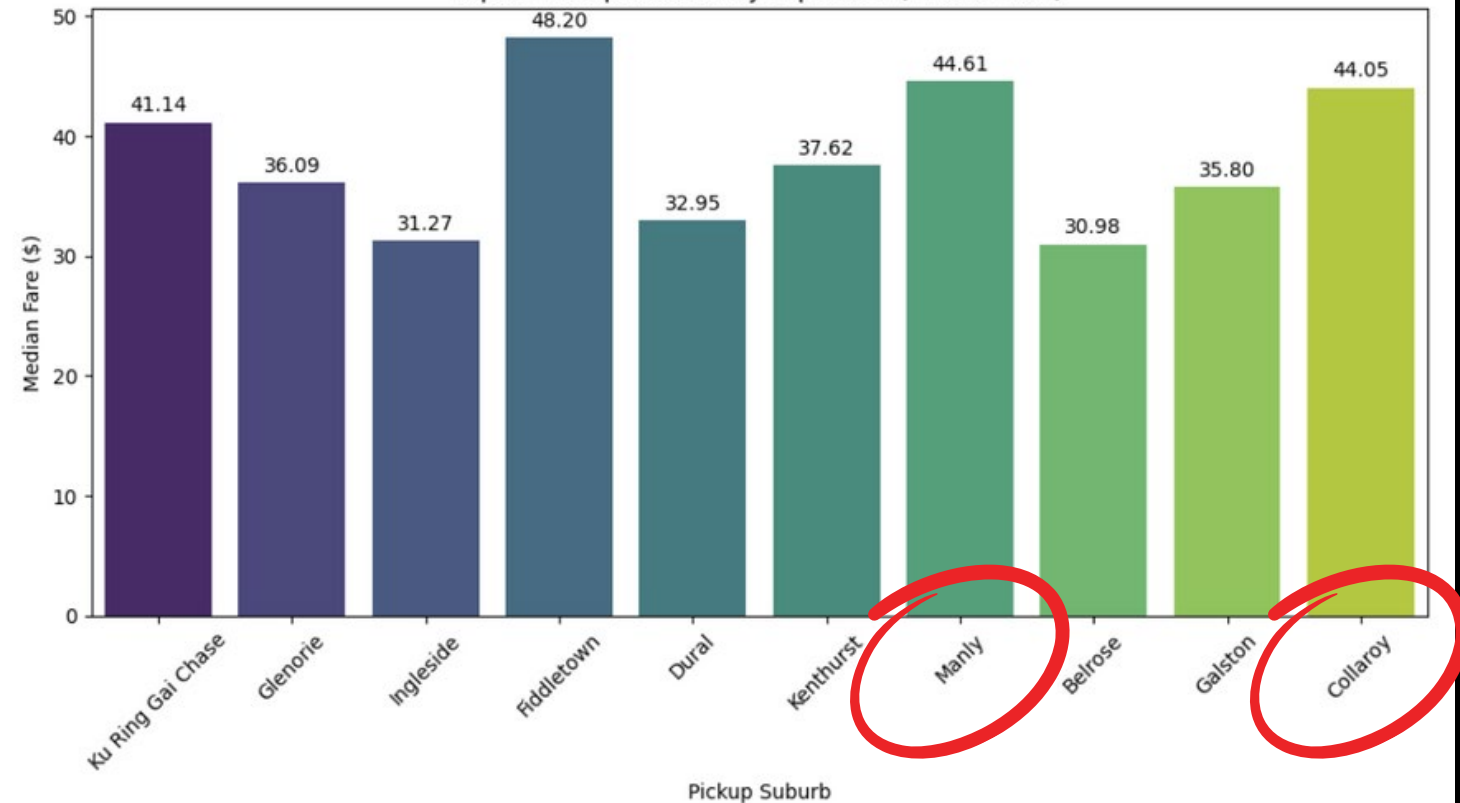
- This suburb covers a very large area
- Remote location with poor public transport access
- Tourists or visitors often rely on Uber to return to the city or hotel

BAR CHART

Top 10 Pickup Suburbs by Trip Count (Median Fare)



Top 10 Pickup Suburbs by Trip Count (Median Fare)

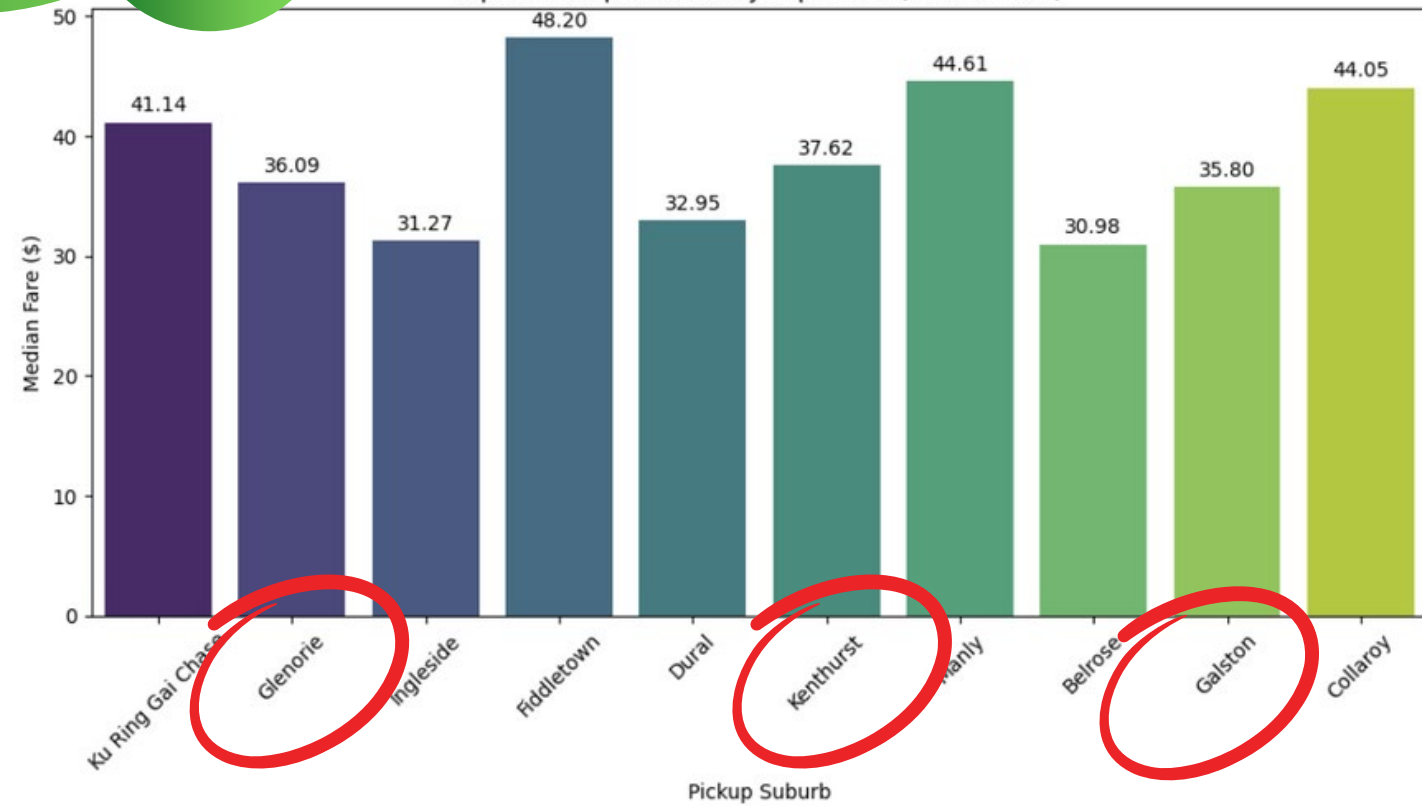


Due to the long distance to the CBD and limited driver supply

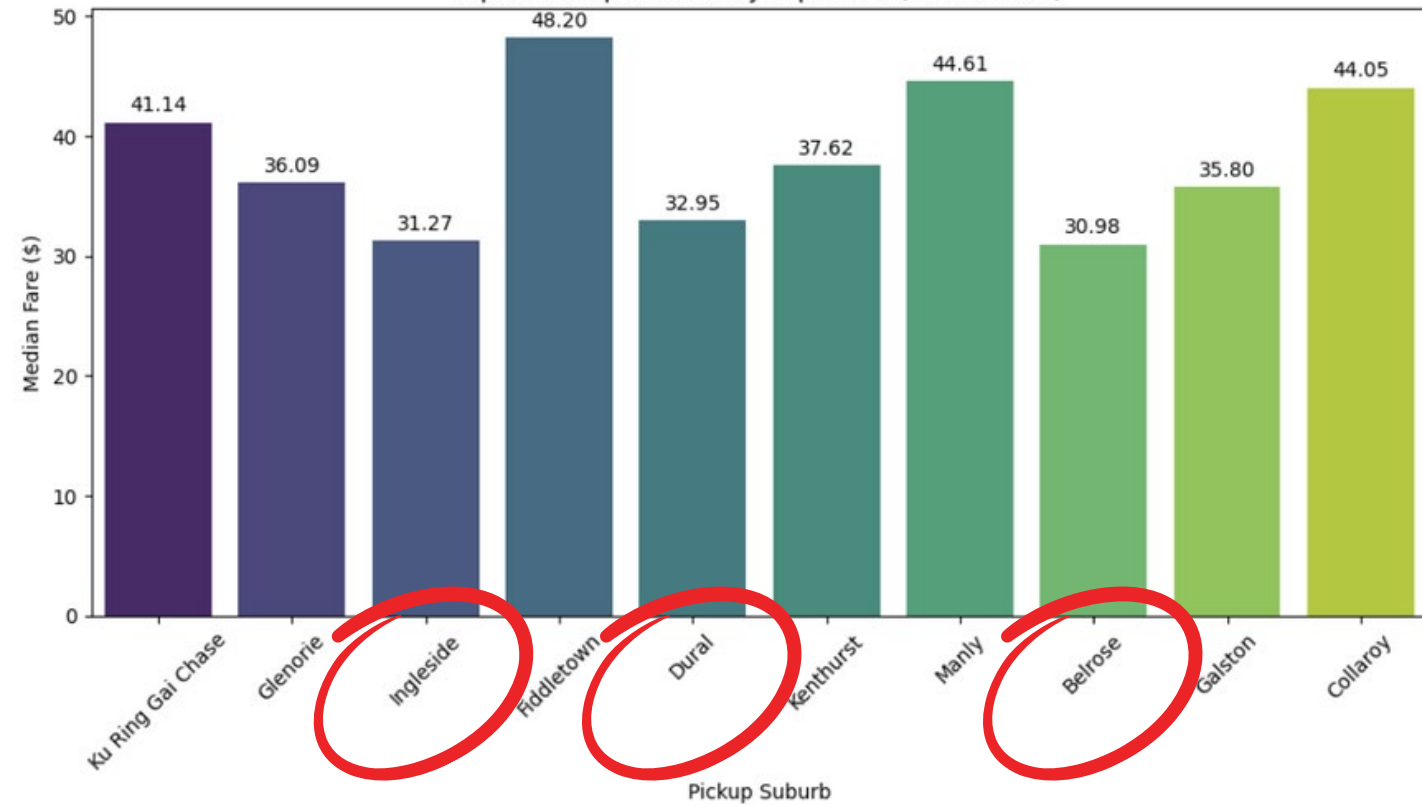
The combination of wealthy residents, frequent tourism, and limited transport alternatives

BAR CHART

Top 10 Pickup Suburbs by Trip Count (Median Fare)



Top 10 Pickup Suburbs by Trip Count (Median Fare)



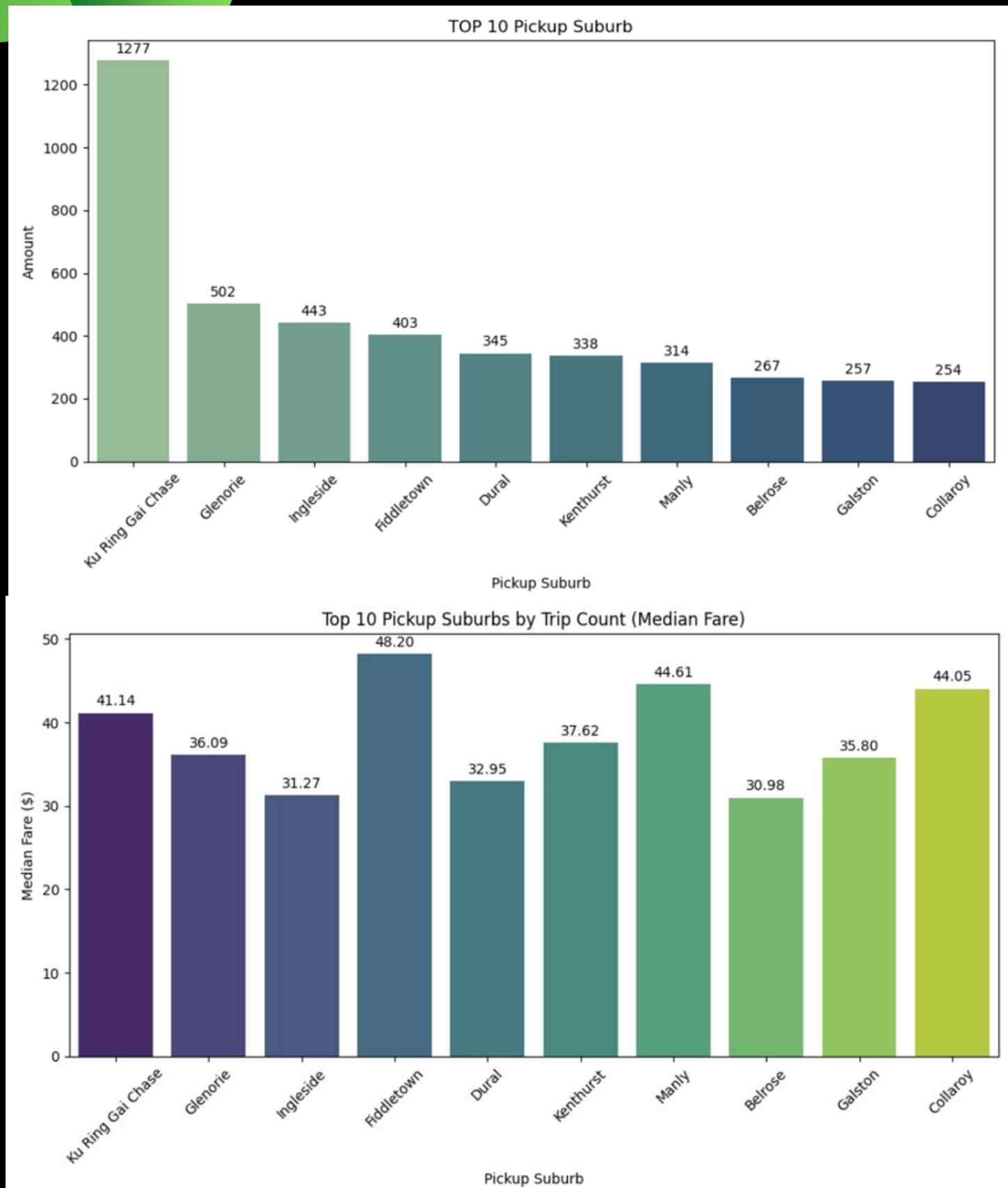
These are fringe suburbs on the edge of the city. Most trips here are likely for daily commuting

These suburbs are closer to main roads and transit routes,

Public transport is more available and competitive,

And most trips are short, everyday local rides taken by residents.

CONCLUSION

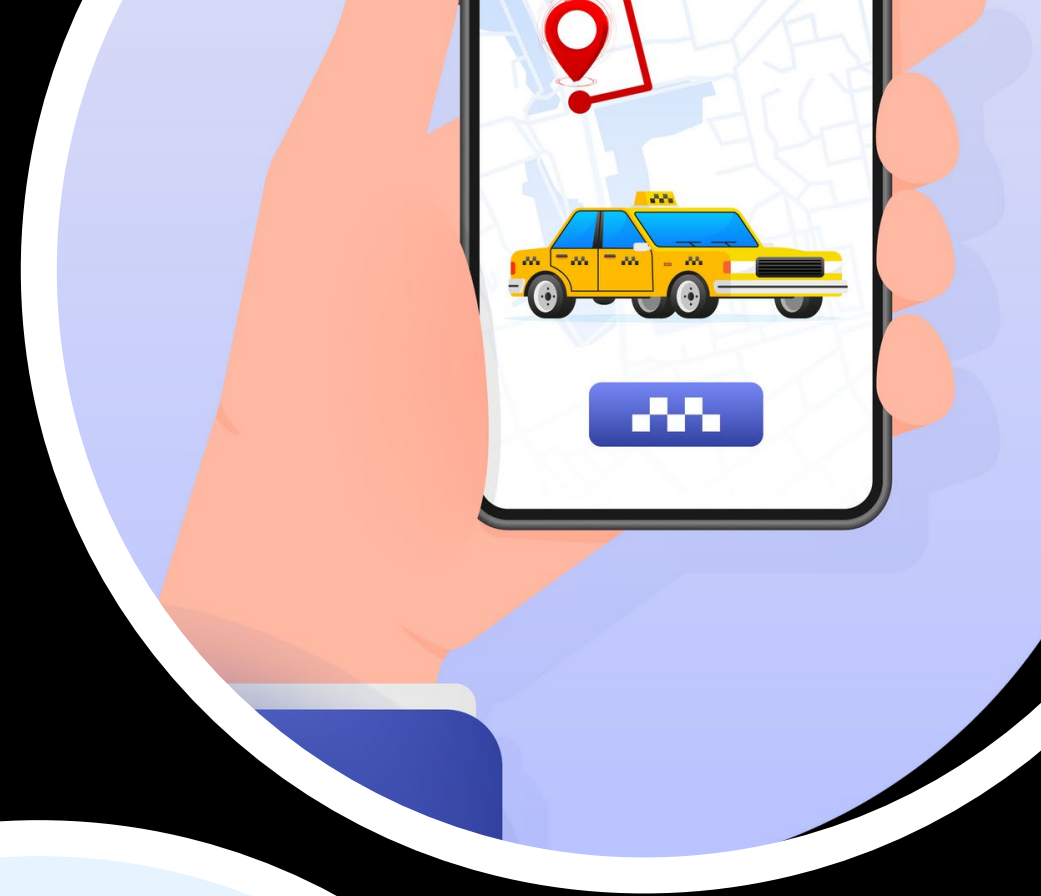


- Geographic location significantly affects ride fare
- Remote areas tend to have higher per prices - ride
- Central areas show price stability despite high demand

DRIVE TIME & DRIVE DISTANCE

● with Scatter Plot and Linear Regression Model

[Read More](#)



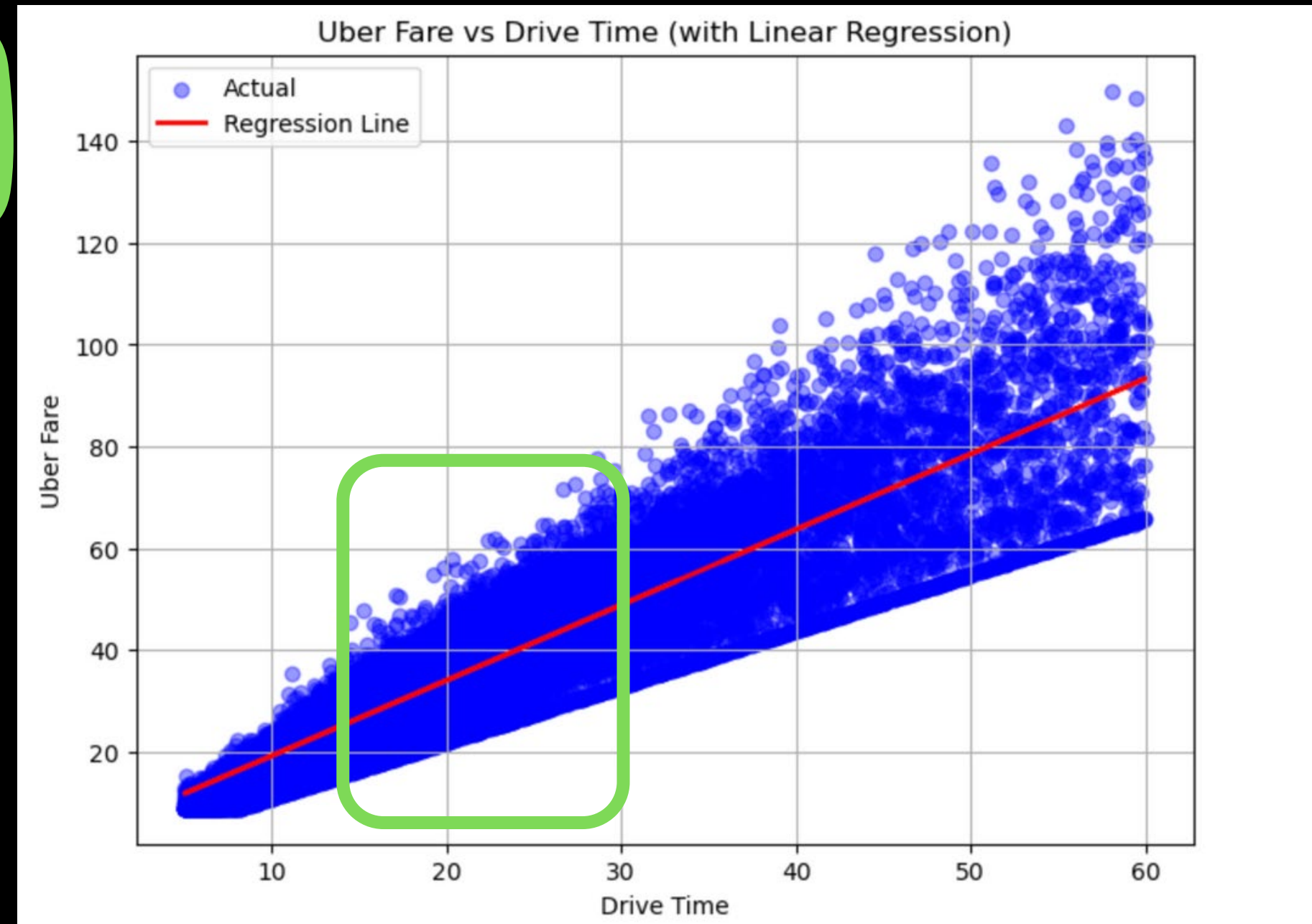
DRIVE TIME -- UBER FARE



$$\text{Uber Fare} = 1.482 * \text{Drive Time} + 4.426$$

$R^2: 0.836$

- For every **1 minute** increase in driving time, the uber fare will increase **\$1.482**

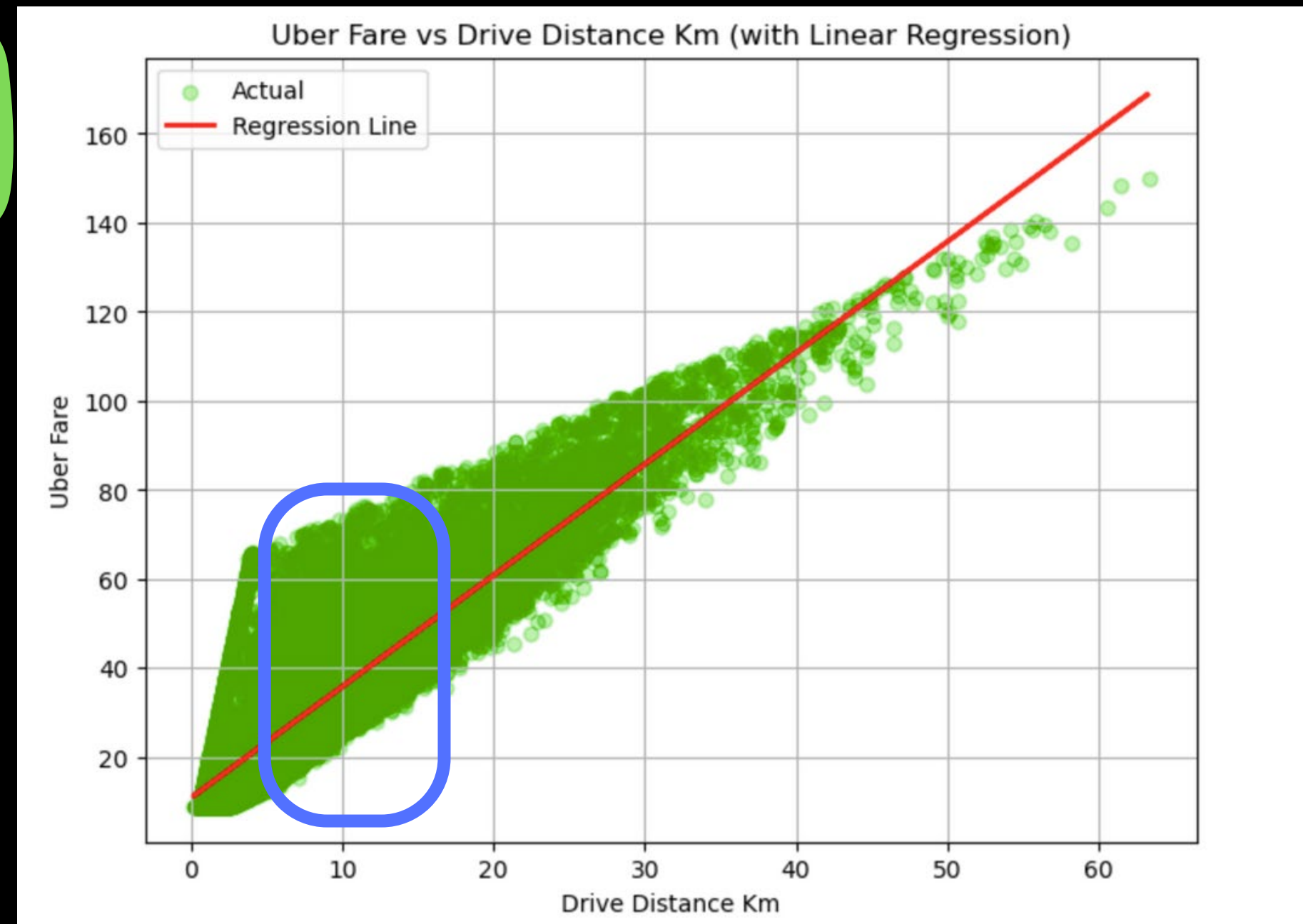


DRIVE DISTANCE — UBER FARE

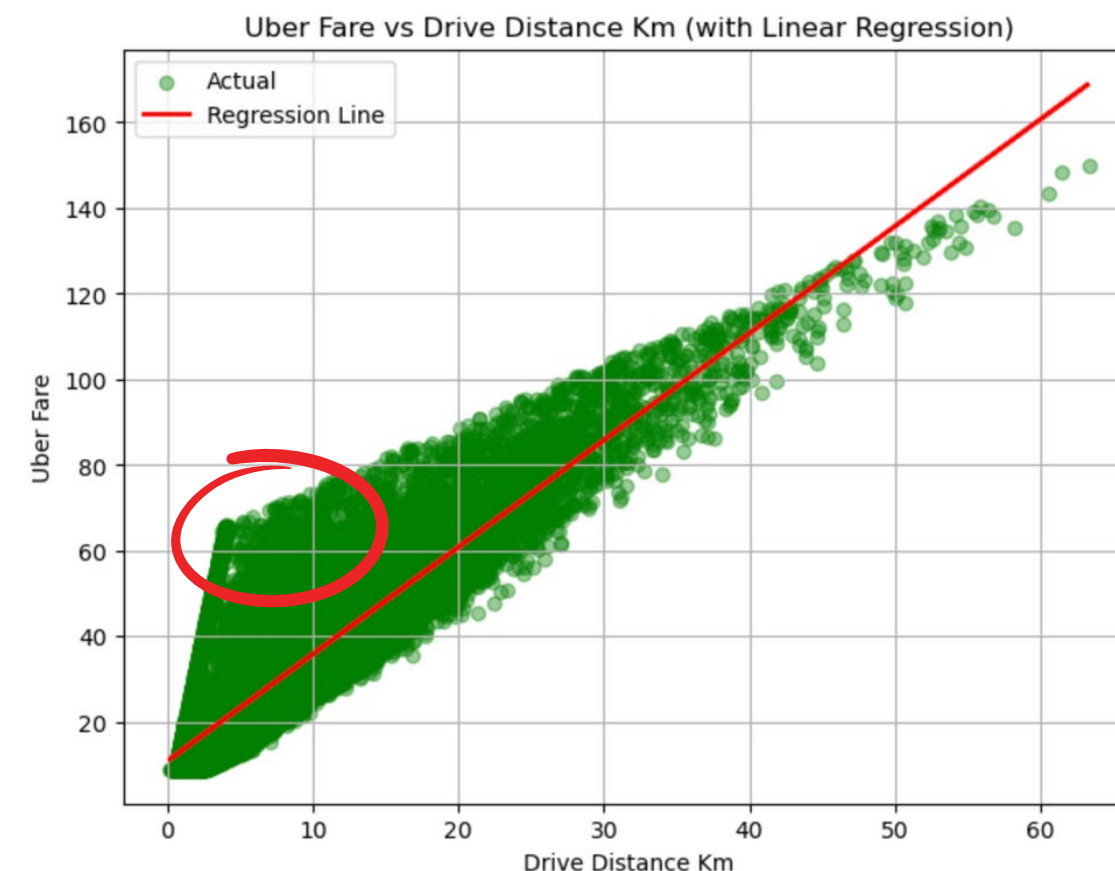
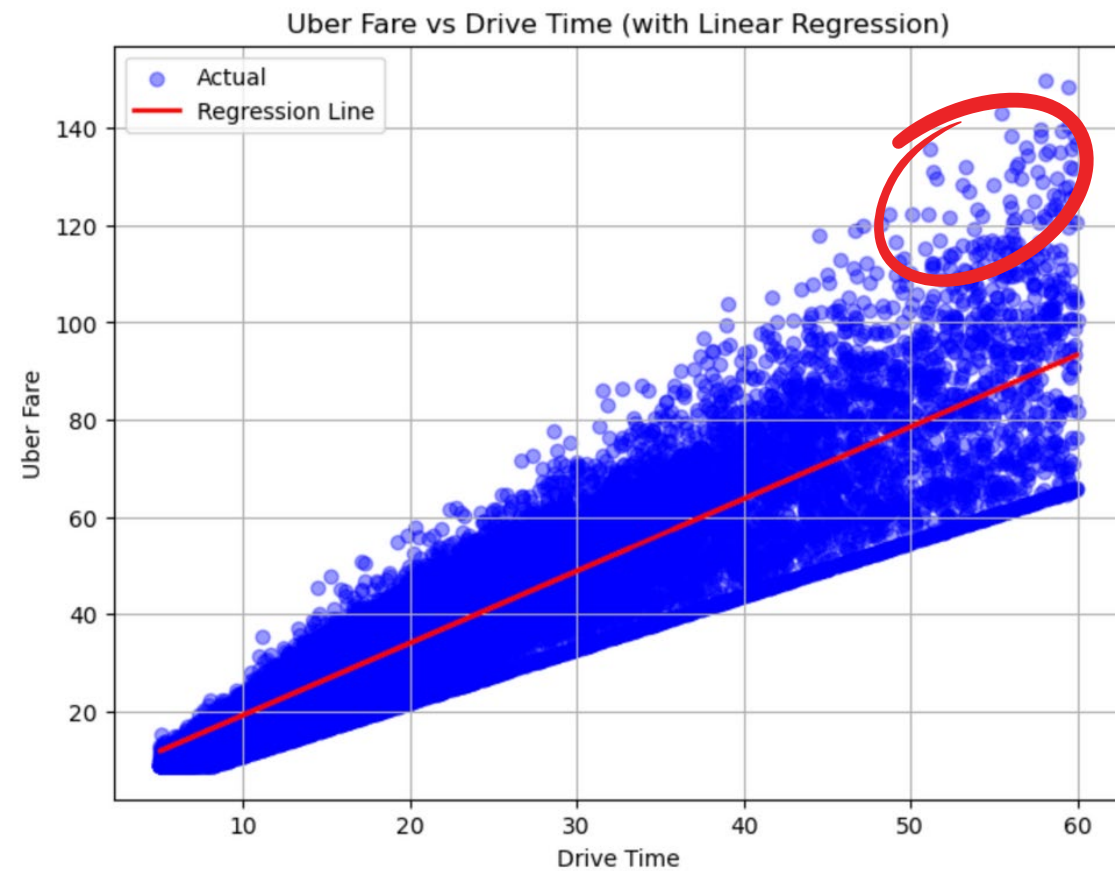
$$\text{Uber Fare} = 2.497 * \text{Drive Distance} + 10.908$$

$R^2: 0.754$

- For every **1 km** increase in driving distance, the uber fare will increase **\$2.497**



EXPLANATIONS OF OUTLIES



- **Rush Hour (surge pricing)**

School zone/ Mid night

- **Busy Traffic & Extreme Weather**

Drive Time increases and the price will increase.

- **Pick -up Area**

More ubers are available in central areas, which is cheaper than in the suburbs

- **Type of Cars (potential)**

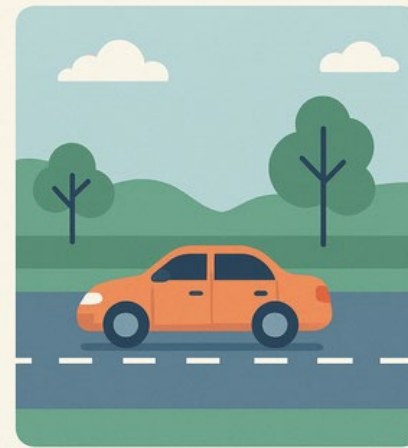
Uber green is cheaper than gasoline cars.



URBAN



SUBURBAN



Choose a ride

Rides we think you'll take

	UberX 4 16:59 · 1 min away Affordable rides, all to yourself	\$20.55
	Comfort Electric 4 17:01 · 3 min away Comfortable fully electric vehicles	\$25.97
	Pool 1 17:03-17:09 · 3 min away Save by sharing	\$16.27
	UberXL 6 17:02 · 4 min away Affordable rides for groups up to 6	\$31.26

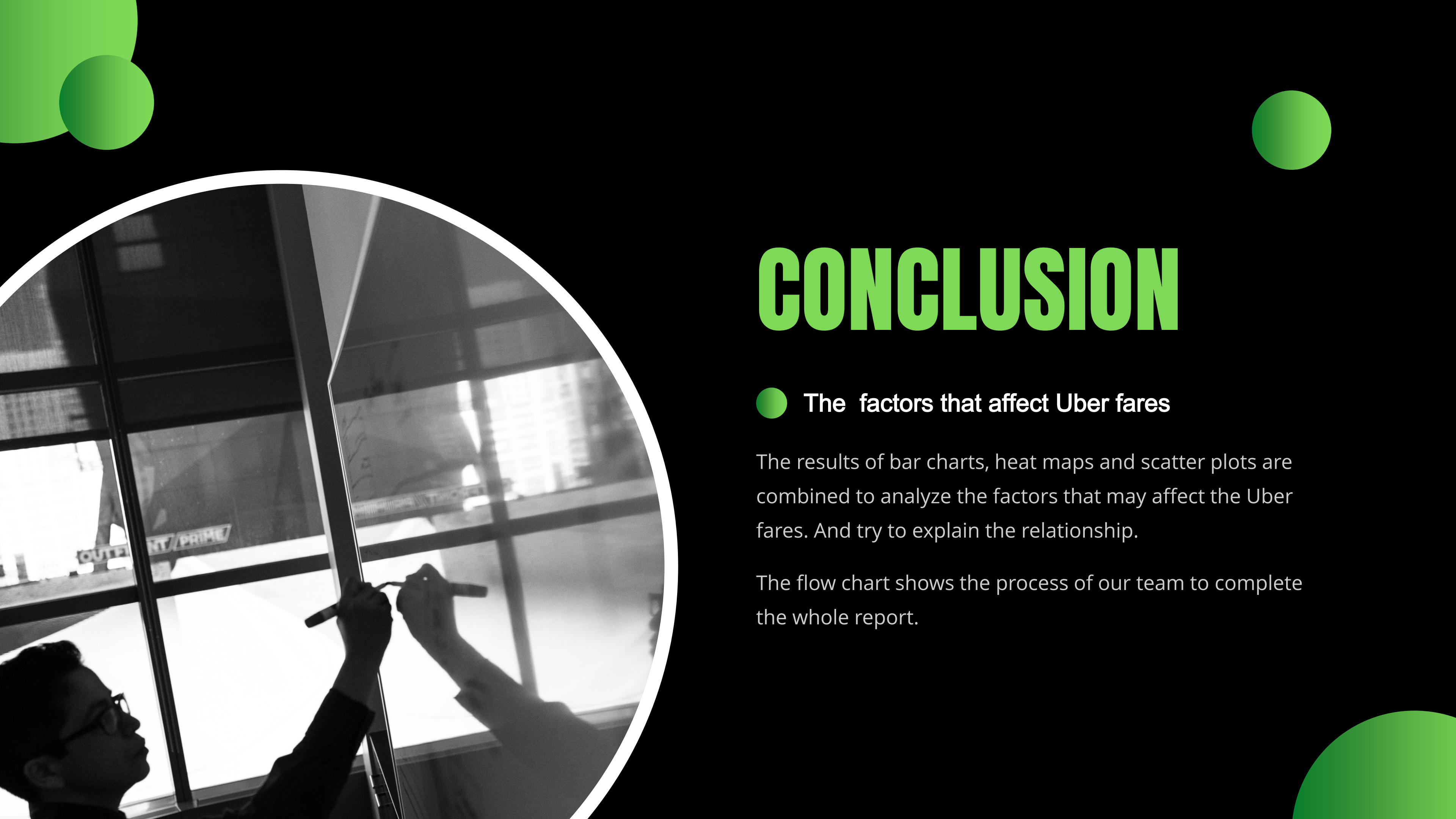
LINEAR REGRESSION MODEL --2 FACTORS

$$\text{Uber Fare} = 0.999 * \text{Drive Time} + 1.447 * \text{Drive Distance} + 0.078$$

$$R^2 : 0.9999$$

when fixed other variable:

- For every 1 minute increase in drive time, the uber fare will increase \$0.999
- For every 1 km increase in drive distance, the uber fare will increase \$1.447
- Drive distance has more significant impact on Uber Fare.



CONCLUSION

● The factors that affect Uber fares

The results of bar charts, heat maps and scatter plots are combined to analyze the factors that may affect the Uber fares. And try to explain the relationship.

The flow chart shows the process of our team to complete the whole report.

THE FACTORS THAT AFFECT UBER FARES

● Peak demand periods

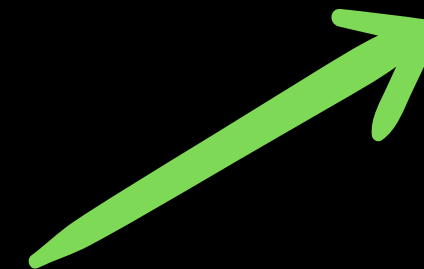
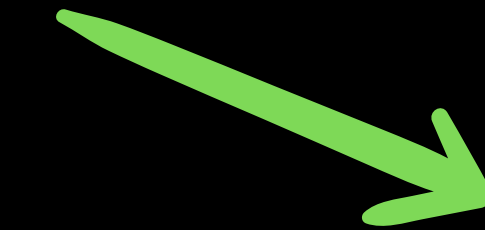
Fares are higher during school zones and peak commuting hours, as shown through the bar chart of median fares in different time periods.

● Concentrated locations

The heat map and regional fare chart show that prices tend to be higher in areas with low supply, while they remain standard in high-demand areas. This indicates a supply-demand imbalance impacting fare levels.

● Distance & Duration

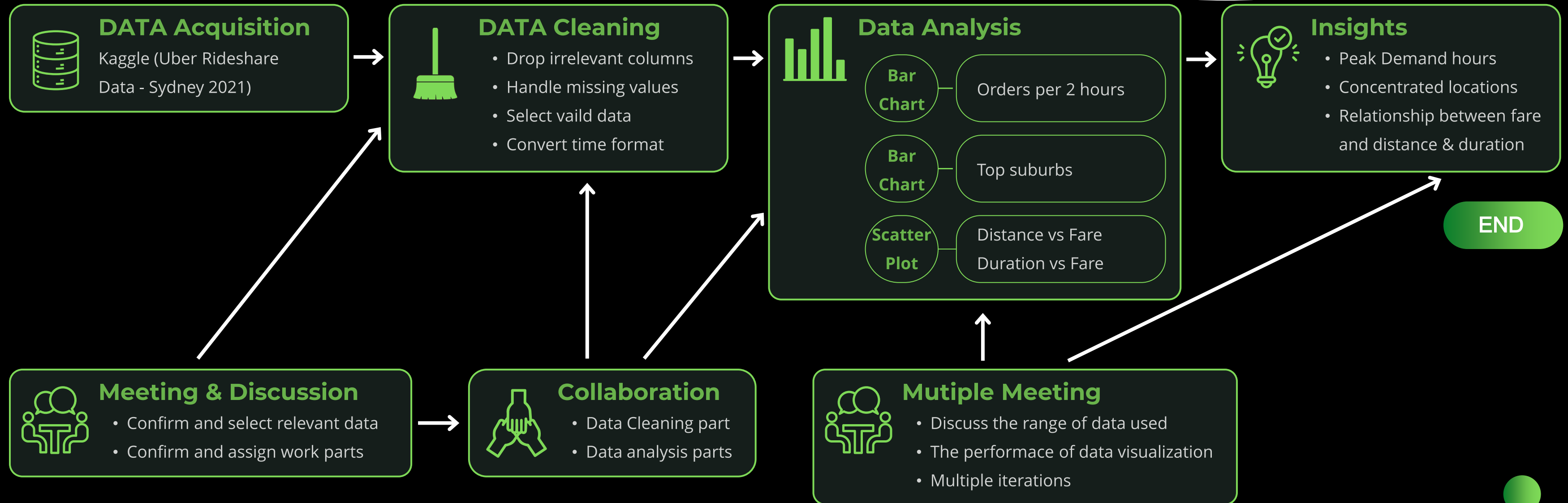
Through scatter plots and regression analysis, we discovered that both order distance and time significantly impact fares—with distance having a greater effect overall.

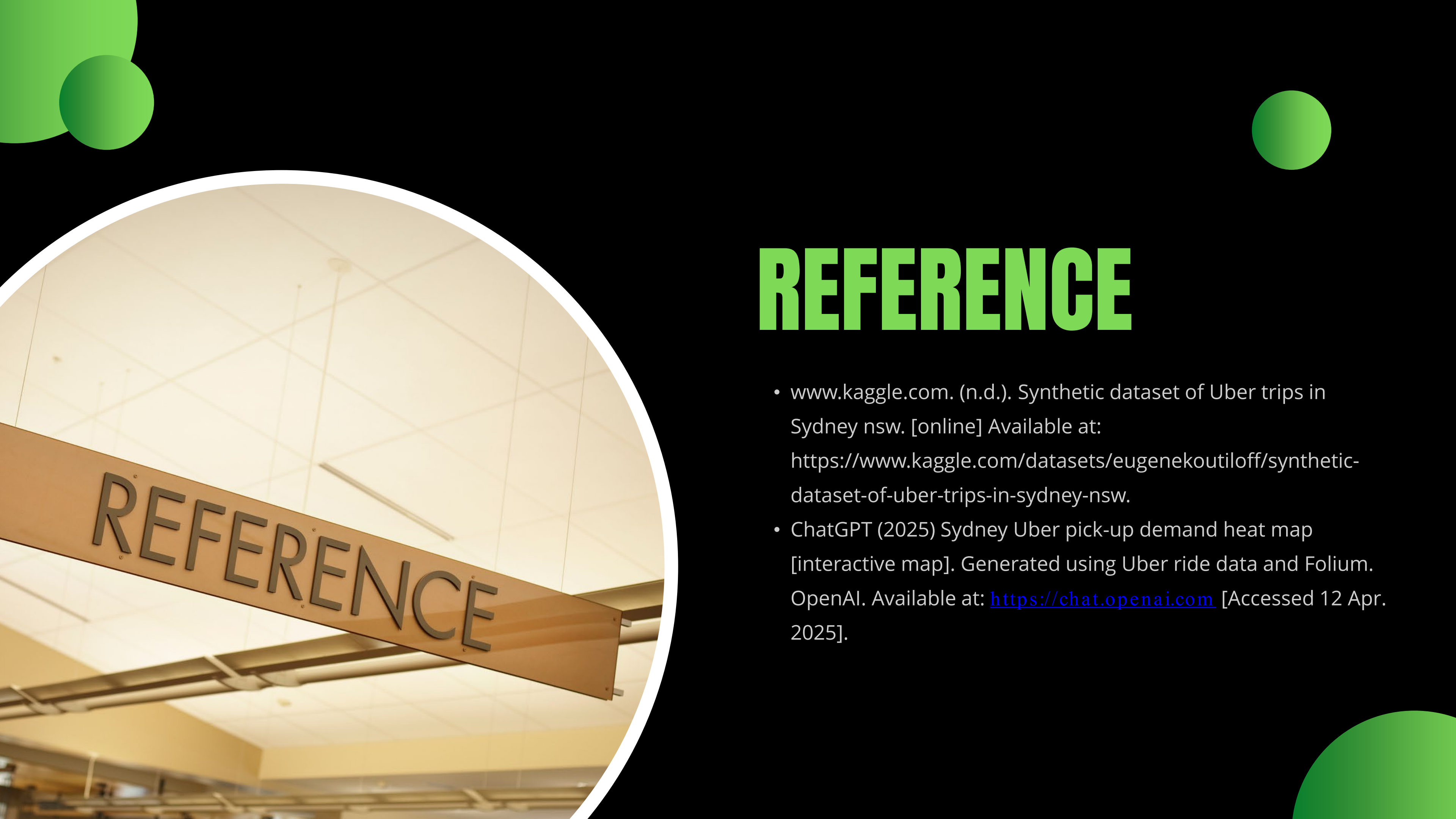


**Flexible
Dynamic
Pricing**

FLOW CHART

START





REFERENCE

- [www.kaggle.com](https://www.kaggle.com/datasets/eugenekoutiloff/synthetic-dataset-of-uber-trips-in-sydney-nsw). (n.d.). Synthetic dataset of Uber trips in Sydney nsw. [online] Available at: <https://www.kaggle.com/datasets/eugenekoutiloff/synthetic-dataset-of-uber-trips-in-sydney-nsw>.
- ChatGPT (2025) Sydney Uber pick-up demand heat map [interactive map]. Generated using Uber ride data and Folium. OpenAI. Available at: <https://chat.openai.com> [Accessed 12 Apr. 2025].

See You Next

THANK YOU